

Advanced Counting Techniques

Chapter 8

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Chapter Summary

- Applications of Recurrence Relations
- Solving Linear Recurrence Relations
 - Homogeneous Recurrence Relations
 - Nonhomogeneous Recurrence Relations
- Divide-and-Conquer Algorithms and Recurrence Relations
- Generating Functions
- Inclusion-Exclusion
- Applications of Inclusion-Exclusion

Applications of Recurrence Relations

Section 8.1

Recurrence Relations

(recalling definitions from Chapter 2)

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that *expresses a_n in terms of one or more of the previous terms* of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the *first term* where the recurrence relation takes effect.

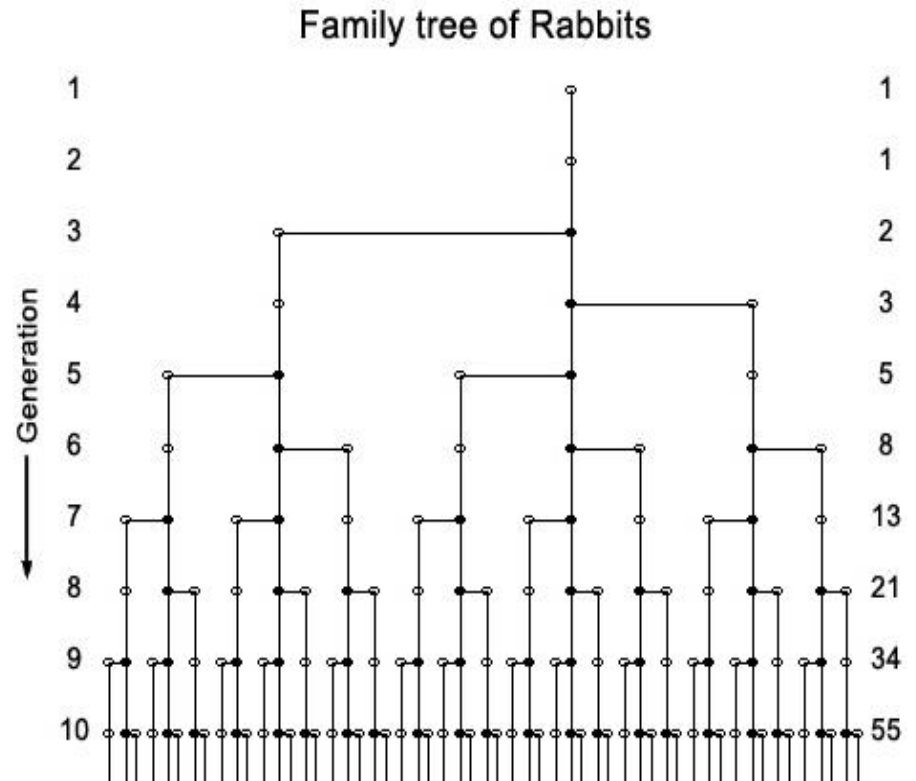
Rabbits and the Fibonacci Numbers

Example: A **young pair of rabbits** (one of each gender) is placed on an island. A pair of rabbits does not breed until they are 2 months old. **After** they are **2 months old**, each pair of rabbits **produces another pair each month**. **Find a recurrence relation for the number of pairs of rabbits on the island at the n -th months,** assuming that rabbits never die.

This is the original problem considered by Leonardo Pisano (Fibonacci) in the thirteenth century.

Rabbits and the Fibonacci Numbers (*cont.*)

Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	Reproducing pairs	Young pairs	Total pairs
		1	0	1	1
		2	0	1	1
		3	1	1	2
		4	1	2	3
		5	2	3	5
		6	3	5	8



Modeling the Population Growth of Rabbits on an Island

Rabbits and the Fibonacci Numbers (*cont.*)

Solution: Let f_n be the number of pairs of rabbits after n months.

- There are $f_1 = 1$ pairs of rabbits on the island at the end of the first month.
- We also have $f_2 = 1$ because the pair does not breed during the first month.
- To find the number of pairs on the island after n months, add the number on the island after the previous month, f_{n-1} , and the number of newborn pairs, which equals f_{n-2} , because each newborn pair comes from a pair at least two months old.

Consequently the sequence $\{f_n\}$ satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$ with the initial conditions $f_1 = 1$ and $f_2 = 1$. The number of pairs of rabbits on the island after n months is given by the n th Fibonacci number.

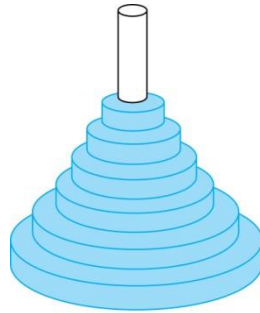
The Tower of Hanoi

In the late 19th century, the French mathematician Édouard Lucas invented a puzzle consisting of three pegs on a board with disks of different sizes. **Initially all of the disks are on the first peg in order of size**, with the largest on the bottom.

Rules: You are allowed to move the disks **one at a time** from one peg to another as long as a **larger disk is never placed on a smaller**.

Goal: Using allowable moves, **end up with all the disks on the second peg** in order of size with largest on the bottom.

The Tower of Hanoi (*continued*)



Peg 1



Peg 2

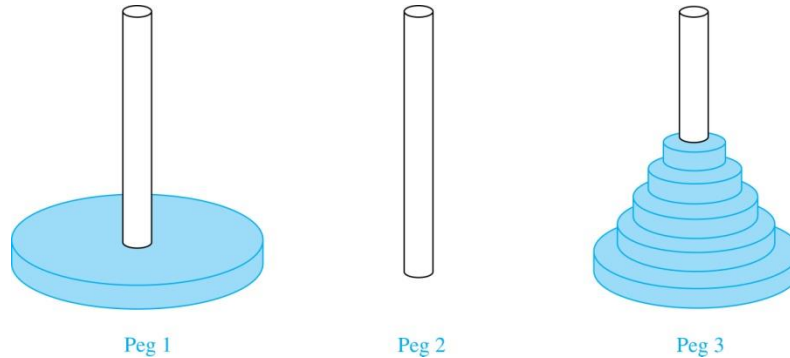


Peg 3

The Initial Position in the Tower of Hanoi Puzzle

The Tower of Hanoi (*continued*)

Solution: Let $\{H_n\}$ denote the number of moves needed to solve the Tower of Hanoi Puzzle with n disks. Set up a recurrence relation for the sequence $\{H_n\}$. Begin with n disks on peg 1. We can transfer the top $n - 1$ disks, following the rules of the puzzle, to peg 3 using H_{n-1} moves.



First, we use 1 move to transfer the largest disk to the second peg. Then we transfer the $n - 1$ disks from peg 3 to peg 2 using H_{n-1} additional moves. This can not be done in fewer steps. Hence,

$$H_n = 2H_{n-1} + 1.$$

The initial condition is $H_1 = 1$ since a single disk can be transferred from peg 1 to peg 2 in one move.

The Tower of Hanoi (*continued*)

- We can use an iterative approach to solve this recurrence relation by repeatedly expressing H_n in terms of the previous terms of the sequence.

$$\begin{aligned}H_n &= 2H_{n-1} + 1 \\&= 2(2H_{n-2} + 1) + 1 = 2^2 H_{n-2} + 2 + 1 \\&= 2^2(2H_{n-3} + 1) + 2 + 1 = 2^3 H_{n-3} + 2^2 + 2 + 1 \\&\vdots \\&= 2^{n-1}H_1 + 2^{n-2} + 2^{n-3} + \dots + 2 + 1 \\&= 2^{n-1} + 2^{n-2} + 2^{n-3} + \dots + 2 + 1 \quad \text{because } H_1 = 1 \\&= 2^n - 1 \quad \text{using the formula for the sum of the terms of a geometric series}\end{aligned}$$

- There was a myth created with the puzzle. Monks in a tower in Hanoi are transferring 64 gold disks from one peg to another following the rules of the puzzle. They move one disk each day. When the puzzle is finished, the world will end.
- Using this formula for the 64 gold disks of the myth,

$$2^{64} - 1 = 18,446, 744,073, 709,551,615$$

days are needed to solve the puzzle, which is more than 500 billion years.

Counting Bit Strings

Example 3: Find a **recurrence relation** and give initial conditions **for the number of bit strings of length n without two consecutive 0s**. How many such bit strings are there of length 5?

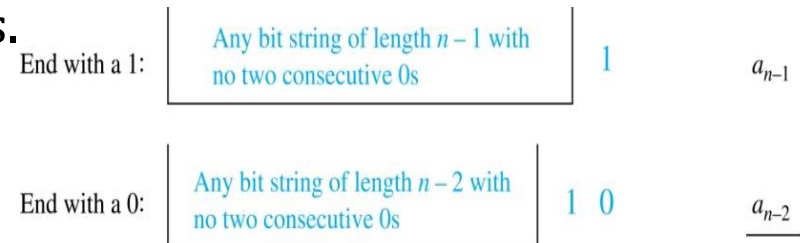
Solution:

Let a_n denote the number of bit strings of length n without two consecutive 0s.

To obtain a recurrence relation for $\{a_n\}$ note **by the sum rule** that a_n is equal to the number of such bit strings ending with a 0 **plus** the number of such bit strings ending with a 1.

Now assume that $n \geq 3$.

– The bit strings of **length n** ending with 1 without two consecutive 0s are the bit strings of length $n-1$ with no two consecutive 0s with a 1 added at the end. Hence, there are a_{n-1} such bit strings.



– The bit strings of length n ending with 0 without two consecutive 0s are the bit strings of length $n-2$ with no two consecutive 0s with 10 at the end. Hence, there are a_{n-2} such bit strings.

We conclude that $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$.

Bit Strings (*continued*)

The **initial conditions** are:

- $a_1 = 2$, since both the bit strings **0** and **1** do not have consecutive 0s.
- $a_2 = 3$, since the bit strings **01**, **10**, and **11** do not have consecutive 0s, while 00 does.

To obtain a_5 , we use the recurrence relation three times to find that:

- $a_3 = a_2 + a_1 = 3 + 2 = 5$
- $a_4 = a_3 + a_2 = 5 + 3 = 8$
- $a_5 = a_4 + a_3 = 8 + 5 = 13$

Note that $\{a_n\}$ satisfies the same recurrence relation as the Fibonacci sequence. Since $a_1 = f_3$ and $a_2 = f_4$, we conclude that $a_n = f_{n+2}$.

Counting the Ways to Parenthesize a Product

Example: Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n + 1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, to specify the order of multiplication.

For example, $C_3 = 5$, since all the possible ways to parenthesize 4 numbers are

$((x_0 \cdot x_1) \cdot x_2) \cdot x_3$, $(x_0 \cdot (x_1 \cdot x_2)) \cdot x_3$, $(x_0 \cdot x_1) \cdot (x_2 \cdot x_3)$, $x_0 \cdot ((x_1 \cdot x_2) \cdot x_3)$, $x_0 \cdot (x_1 \cdot (x_2 \cdot x_3))$

Solution:

Note that however parentheses are inserted in $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, one “ $\cdot$ ” operator remains outside all parentheses.

This final operator appears between two of the $n + 1$ numbers, say x_k and x_{k+1} .

Since there are C_k ways to insert parentheses in the product $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_k$ and C_{n-k-1} ways to insert parentheses in the product $x_{k+1} \cdot x_{k+2} \cdot \cdots \cdot x_n$, we have

$$\begin{aligned} C_n &= C_0 C_{n-1} + C_1 C_{n-2} + \cdots + C_{n-2} C_1 + C_{n-1} C_0 \\ &= \sum_{k=0}^{n-1} C_k C_{n-k-1} \end{aligned}$$

The initial conditions are $C_0 = 1$ and $C_1 = 1$.

The sequence $\{C_n\}$ is the sequence of **Catalan Numbers**. This recurrence relation can be solved using the method of **generating functions** (will be introduced in Section 8.4)

Solving Linear Recurrence Relations

Section 8.2

Linear Homogeneous Recurrence Relations

Definition: A *linear*(線性) *homogeneous*(齊次) recurrence relation of degree k with constant coefficients is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$, where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$

- it is *linear* because the right-hand side is a sum of the previous terms of the sequence each multiplied by a constant.
- it is *homogeneous* because no terms occur that are not multiples of the a_j s. Each coefficient is a constant.
- the *degree is k* because a_n is expressed in terms of the *previous k terms of the sequence*.

*Note: A sequence satisfying such a recurrence relation is uniquely determined by the recurrence relation and the *k initial conditions* $a_0 = C_1, a_1 = C_2, \dots, a_{k-1} = C_k$.

Examples of Linear Homogeneous Recurrence Relations

- $P_n = (1.11)P_{n-1}$ linear homogeneous recurrence relation of degree one
- $f_n = f_{n-1} + f_{n-2}$ linear homogeneous recurrence relation of degree two
- $a_n = a_{n-1} + a_{n-2}^2$ not linear
- $H_n = 2H_{n-1} + 1$ not homogeneous
- $B_n = nB_{n-1}$ coefficients are not constants

Solving Linear Homogeneous Recurrence Relations

- The basic approach is to look for solutions of the form $a_n = r^n$, where r is a constant.
- Note that $a_n = r^n$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ iff $r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}$.
- Algebraic manipulation yields the *characteristic equation* (特徵方程式): $r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$
- The sequence $\{a_n\}$ with $a_n = r^n$ is a solution iff r is a solution to the characteristic equation.
- The solutions to the characteristic equation are called the *characteristic roots* (特徵根) of the recurrence relation. The roots are used to give an explicit formula for all the solutions of the recurrence relation.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Recall: Linear Homogeneous R.R. of Degree Two: $a_n = c_1 a_{n-1} + c_2 a_{n-2}$

Theorem 1: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Proof:

($\leftarrow$) (i.e., $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n \rightarrow a_n = c_1 a_{n-1} + c_2 a_{n-2}$)

$$\begin{aligned} c_1 a_{n-1} + c_2 a_{n-2} &= c_1 (\alpha_1 r_1^{n-1} + \alpha_2 r_2^{n-1}) + c_2 (\alpha_1 r_1^{n-2} + \alpha_2 r_2^{n-2}) \\ &= \alpha_1 r_1^{n-2} (c_1 r_1 + c_2) + \alpha_2 r_2^{n-2} (c_1 r_2 + c_2) = \alpha_1 r_1^n + \alpha_2 r_2^n = a_n \end{aligned}$$

($\rightarrow$) (i.e., $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ with $a_0 = K_0$ and $a_1 = K_1 \rightarrow a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$)

We will prove this by using **strong induction**:

Basis Step:

- $a_0 = K_0 = \alpha_1 r_1^0 + \alpha_2 r_2^0 = \alpha_1 + \alpha_2$;
- $a_1 = K_1 = \alpha_1 r_1 + \alpha_2 r_2$ (since $r_1 \neq r_2$, (α_1, α_2) must exist!) ,

Induction Step:

Inductive hypothesis: $a_j = \alpha_1 r_1^j + \alpha_2 r_2^j$, for $1 \leq j \leq k$

$$\begin{aligned} a_{k+1} &= c_1 a_k + c_2 a_{k-1} \\ &= (\text{by ind.hyp.}) c_1 (\alpha_1 r_1^k + \alpha_2 r_2^k) + c_2 (\alpha_1 r_1^{k-1} + \alpha_2 r_2^{k-1}) \\ &= \alpha_1 (c_1 r_1^k + c_2 r_1^{k-1}) + \alpha_2 (c_1 r_2^k + c_2 r_2^{k-1}) \\ &= \alpha_1 r_1^{k-1} (c_1 r_1 + c_2) + \alpha_2 r_2^{k-1} (c_1 r_2 + c_2) = \alpha_1 r_1^{k+1} + \alpha_2 r_2^{k+1} \end{aligned}$$

based on mathematical induction have shown that $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, for $n = 0, 1, 2, \dots$

Using Theorem 1

Example: What is the solution to the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0 = 2 \text{ and } a_1 = 7?$$

Solution: The **characteristic equation** is $r^2 - r - 2 = 0$.

Its roots are $r = 2$ and $r = -1$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, for some constants α_1 and α_2 .

To find the constants α_1 and α_2 , note that

$$a_0 = 2 = \alpha_1 + \alpha_2 \text{ and } a_1 = 7 = \alpha_1 2 + \alpha_2 (-1).$$

Solving these equations, we find that $\alpha_1 = 3$ and $\alpha_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

An Explicit Formula for the Fibonacci Numbers

We can use Theorem 1 to find an explicit formula for the Fibonacci numbers. The sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

Solution: The roots of the characteristic equation $r^2 - r - 1 = 0$ are

$$r_1 = \frac{1+\sqrt{5}}{2}$$

$$r_2 = \frac{1-\sqrt{5}}{2}$$

Fibonacci Numbers (*continued*)

Therefore by Theorem 1

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

for some constants α_1 and α_2 .

Using the initial conditions $f_0 = 0$ and $f_1 = 1$, we have

$$f_0 = \alpha_1 + \alpha_2 = 0$$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1 .$$

$$\alpha_1 = \frac{1}{\sqrt{5}} , \quad \alpha_2 = -\frac{1}{\sqrt{5}}$$

Solving, we obtain

Hence,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

The Solution when there is a **Repeated Root**(重根)

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has **only one root** r_0 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Proof: $\Rightarrow$ (Similar to Theorem 1; Reading Exercise!)

($\leftarrow$): (i.e., $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n \rightarrow a_n = c_1 a_{n-1} + c_2 a_{n-2}$)

$$\begin{aligned} c_1 a_{n-1} + c_2 a_{n-2} &= c_1 (\alpha_1 r_0^{n-1} + \alpha_2 (n-1) r_0^{n-1}) + c_2 (\alpha_1 r_0^{n-2} + \alpha_2 (n-2) r_0^{n-2}) \\ &= \alpha_1 r_0^{n-2} (c_1 r_0 + c_2) + \alpha_2 r_0^{n-2} ((n-1)(c_1 r_0 + c_2) - c_2) \\ &= \alpha_1 r_0^n + \alpha_2 r_0^{n-2} ((n-1) r_0^2 + r_0^2) = \alpha_1 r_0^n + \alpha_2 n r_0^n = a_n \end{aligned}$$

$$\begin{aligned} \because r^2 - c_1 r - c_2 &= (r - r_0)^2 \\ &= r^2 - 2 r_0 r + r_0^2; \quad \Rightarrow -c_2 = r_0^2 \end{aligned}$$

($\rightarrow$): (i.e., $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ with $a_0 = C_0$ and $a_1 = C_1 \rightarrow a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$)

We will prove this by using **strong induction**:

Basis Step:

- $a_0 = C_0 = \alpha_1 r_0^0 + \alpha_2 \cdot 0 \cdot r_0^0 = \alpha_1$;
- $a_1 = C_1 = \alpha_1 r_0 + \alpha_2 \cdot 1 \cdot r_0$ (such (α_1, α_2) exist! $(\alpha_1, \alpha_2) = (C_0, -C_0 + C_1/r_0)$)

Induction Step:

Inductive hypothesis: $a_j = \alpha_1 r_0^j + \alpha_2 j r_0^j$, for $1 \leq j \leq k$

$$\begin{aligned} a_n &= c_1 a_{n-1} + c_2 a_{n-2} = (\text{by ind.hyp.}) c_1 (\alpha_1 r_0^{n-1} + \alpha_2 (n-1) r_0^{n-1}) + c_2 (\alpha_1 r_0^{n-2} + \alpha_2 (n-2) r_0^{n-2}) \\ &= \alpha_1 (c_1 r_0^{n-1} + c_2 r_0^{n-2}) + \alpha_2 (c_1 (n-1) r_0^{n-1} + c_2 (n-2) r_0^{n-2}) \\ &= \alpha_1 r_0^{n-2} (c_1 r_0 + c_2) + \alpha_2 r_0^{n-2} ((n-1)(c_1 r_0 + c_2) - c_2) = \alpha_1 r_0^n + \alpha_2 r_0^{n-2} ((n-1) r_0^2 + r_0^2) \\ &= \alpha_1 r_0^n + \alpha_2 n r_0^n \end{aligned}$$

based on mathematical induction have shown that $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for $n=0, 1, 2, \dots$

Using Theorem 2

Example: What is the solution to the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$?

Solution: The characteristic equation is $r^2 - 6r + 9 = 0$.

The only root is $r = 3$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if

$$a_n = \alpha_1 3^n + \alpha_2 n(3)^n$$

where α_1 and α_2 are constants.

To find the constants α_1 and α_2 , note that

$$a_0 = 1 = \alpha_1 \quad \text{and} \quad a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3.$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Hence,

$$a_n = 3^n + n3^n.$$

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

Theorem 3: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation $r^k - c_1 r^{k-1} - \dots - c_k = 0$ has k distinct roots $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$, for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example 6: Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0=2$ and $a_1=5$, and $a_2=15$.

Solution: (show time)

The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation $r^k - c_1 r^{k-1} - \dots - c_k = 0$ has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$, respectively so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Example 8: Find the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with the initial conditions $a_0=1$ and $a_1=-2$, and $a_2=-1$.

Solution: (show time)

Linear **Non**homogeneous Recurrence Relations **with Constant Coefficients**

Definition: A linear *nonhomogeneous* recurrence relation with constant coefficients is a recurrence relation of the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $F(n)$ is a function not identically zero depending only on n .

The recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

is called the **associated homogeneous recurrence relation**.

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients (*cont.*)

- The following are linear nonhomogeneous recurrence relations with constant coefficients:

$$a_n = a_{n-1} + 2^n,$$

$$a_n = a_{n-1} + a_{n-2} + n^2 + n + 1,$$

$$a_n = 3a_{n-1} + n3^n,$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$$

where the following are the associated linear homogeneous recurrence relations, respectively:

$$a_n = a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2},$$

$$a_n = 3a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

Solving Linear **Non**homogeneous Recurrence Relations with **Constant Coefficients**

Theorem 5: If $\{a_n^{(p)}\}$ is a **particular solution** of the nonhomogeneous linear recurrence relation with constant coefficients $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$, then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a **solution of the associated homogeneous recurrence relation** $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$.

Example: Find all solutions of the recurrence relation $a_n = 3 a_{n-1} + 2n$, and what is the solution with $a_1 = 3$?

Sol: The key to the problem is to find a particular solution $a_n^{(p)}$.

We first **guess** that $a_n^{(p)}$ is in linear form, i.e., $a_n^{(p)} = cn + d$ for some constants c and d .

We then have $cn + d = 3(c(n-1) + d) + 2n = (3c + 2)n + (3d - 3c)$.

As a result $(2c+2)n + (2d - 3c) = 0$. Hence $c = -1$ and $d = -3/2$. $\Rightarrow a_n^{(p)} = -n - 3/2$

Next Find general solution of the associated lhrr: $a_n^{(h)} = 3 a_{n-1}$.

Which we know is of the form $\alpha 3^n$. (特徵方程式: $r-3=0$)

Hence by Theorem 5, **all solutions are of the form:** $a_n = \alpha 3^n - n - 3/2$.

If $a_1 = 3$, then $3 = \alpha 3^1 - 1 - 3/2$ and $\alpha = 11/6$.

Hence $a_n = 11/6 \times 3^n - n - 3/2$.

Solving Linear **Non**homogeneous Recurrence Relations with **Constant Coefficients**

- The key to the problem is to find a particular solution p_n ! How?
- Theorem 6:** Suppose $\{a_n\}$ satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$$

where $c_1, c_2, \dots, c_k$ be real numbers and

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \dots + b_1 n + b_0) * s^n$$

where $b_1, b_2, \dots, b_t$ and s are real numbers.

(1) When s is not a **root of the characteristic equation of the associated linear homogeneous recurrence relation**, there is a particular solution of the form

$$(p_1 n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) * s^n$$

(2) When s is a root of this characteristic equation and its **multiplicity is m** , there is a particular solution of the form

$$n^m (p_1 n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) * s^n$$

以前頁Example為例

$$a_n = 3 a_{n-1} + \underline{2n}$$

$$F(n) = (b_1 n + b_0) * s^n$$

2 0 1

$$\text{Lhrr: } a_n = 3 a_{n-1}$$

特徵根: $r=3$

$s=1$ 不是 特徵根

$$a_n^{(p)} = (p_1 n + p_0) * 1^n$$

Recall:

前頁Example我們猜

$$a_n^{(p)} = cn + d$$

Generating Functions

Section 8.4

Generating Functions

Definition: The *generating function* for the sequence $a_0, a_1, \dots, a_k, \dots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \cdots + a_kx^k + \cdots = \sum_{k=0}^{\infty} a_kx^k.$$

Examples:

- The sequence $\{a_k\}$ with $a_k = 3$ has the generating function $\sum_{k=0}^{\infty} 3x^k$.
- The sequence $\{a_k\}$ with $a_k = k + 1$ has the generating function $\sum_{k=0}^{\infty} (k + 1)x^k$.
- The sequence $\{a_k\}$ with $a_k = 2^k$ has the generating function $\sum_{k=0}^{\infty} 2^k x^k$.

Generating Functions for Finite Sequences

- Generating functions for finite sequences of real numbers can be defined by extending a finite sequence $a_0, a_1, \dots, a_n$ into an infinite sequence by setting $a_{n+1}=0, a_{n+2}=0$, and so on.
- The generating function $G(x)$ of this finite sequence $\{a_n\}$ is a polynomial of degree n because no terms of the form $a_j x^j$ with $j > n$ occur, that is,
$$G(x) = a_0 + a_1 x + \dots + a_n x^n.$$

Example: What is the generating function for the sequence 1,1,1,1,1,1?

Solution: The generating function of 1,1,1,1,1,1 is

$$1 + x + x^2 + x^3 + x^4 + x^5.$$

By Theorem 1 of Section 2.4 (等比級數), we have

$$(x^6 - 1)/(x - 1) = 1 + x + x^2 + x^3 + x^4 + x^5 \text{ when } x \neq 1.$$

Consequently $G(x) = (x^6 - 1)/(x - 1)$ is the generating function of the sequence.

Useful Generating Functions

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \cdots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \cdots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \cdots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise

Useful Generating Functions

Reading Exercise: The EXTENDED BINOMIAL THEOREM (Text Book p.566-567)

$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \dots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \dots$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \dots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2 x^2 + \dots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$(-1)^{k+1}/k$

Note: The series for the last two generating functions can be found in most calculus books when power series are discussed.

Counting Problems and Generating Functions

Example: Find the number of solutions of

$$e_1 + e_2 + e_3 = 17,$$

where e_1, e_2 , and e_3 are nonnegative integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

Solution: The number of solutions is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5) (x^3 + x^4 + x^5 + x^6) (x^4 + x^5 + x^6 + x^7).$$

This follows because a term equal to x^{17} is obtained in the product by picking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3} , where $e_1 + e_2 + e_3 = 17$.

There are three solutions since the coefficient of x^{17} in the product is 3.

Counting Problems and Generating Functions (*continued*)

Example: Use generating functions to find the number of k -combinations of a set with n elements, i.e., $C(n,k)$.

Solution:

Each of the n elements in the set contributes the term $(1+x)$ to the generating function $f(x) = \sum_{k=0}^n a^k x^k = (x^0+x^1) (x^0+x^1) \dots (x^0+x^1) = (1+x)^n$.

solu: = coefficient of x^k in the generating function

By Binominal theorem :

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k, \text{ where } \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

$$\text{Hence \#k-combinations} = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Using Generating Functions to Solve Recurrent Relations

Example: Suppose that a valid codeword is an n -digit number in decimal notation containing an even number of 0s. Let a_n denote the number of valid codewords of length n . It follows from Example 4 of Section 8.1 that

$$a_n = 8a_{n-1} + 10^{n-1} \text{ and the initial condition } a_1 = 9.$$

Use **generating function** to find an explicit formula for a_n .

Solution: For simplicity, we *extend this sequence by setting* $a_0 = 1$.

$$a_n = 8a_{n-1} + 10^{n-1} \Rightarrow a_n x^n = 8a_{n-1} x^n + 10^{n-1} x^n$$

$$\text{Let } G(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$G(x) - 1 = \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} x^n + 10^{n-1} x^n) = 8 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n$$

$$= 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1} = 8x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} 10^n x^n = 8xG(x) + \frac{x}{1-10x}$$

$$\Rightarrow G(x) (1-8x) = 1 + x/(1-10x)$$

$$\Rightarrow G(x) = \frac{1-9x}{(1-8x)(1-10x)} = \frac{1}{2} \left(\frac{1}{1-8x} + \frac{1}{1-10x} \right) = \frac{1}{2} \left(\sum_{n=0}^{\infty} 8^n x^n + \sum_{n=0}^{\infty} 10^n x^n \right) = \sum_{n=0}^{\infty} \frac{1}{2} (8^n + 10^n) x^n$$

$$\text{Hence } a_n = \frac{1}{2} (8^n + 10^n).$$

Note: Using the method introduced in 8.2 is easier than using generating function.

Inclusion-Exclusion

Section 8.5

Principle of Inclusion-Exclusion

- In Section 2.2, we developed the following formula for the number of elements in the union of two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

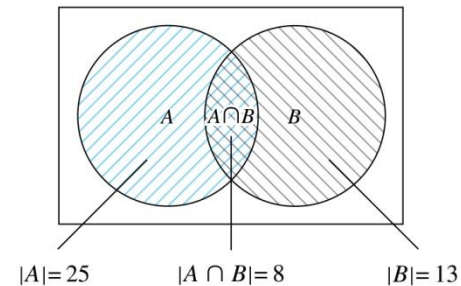
- We will generalize this formula to finite sets of any size.

Two Finite Sets

Example: In a discrete mathematics class every student is a major in computer science or mathematics or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in the class?

Solution: $|A \cup B| = |A| + |B| - |A \cap B|$
 $= 25 + 13 - 8 = 30$

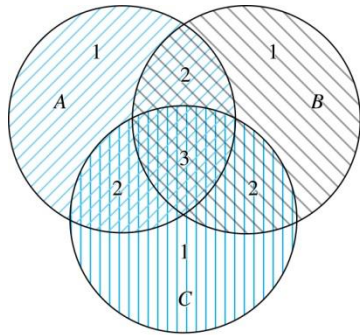
$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$$



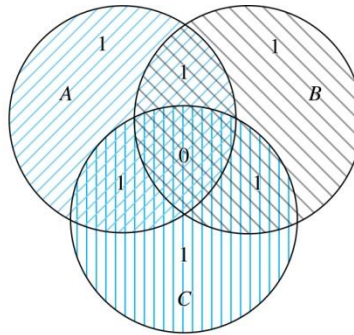
Three Finite Sets

$$|A \cup B \cup C| =$$

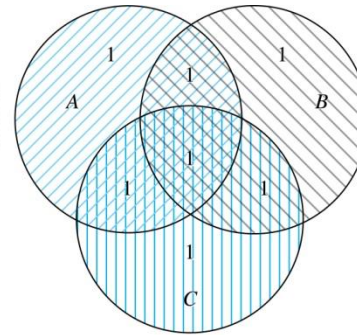
$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



(a) Count of elements by
 $|A| + |B| + |C|$



(b) Count of elements by
 $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C|$



(c) Count of elements by
 $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Three Finite Sets Continued

Example: A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken a course in at least one of Spanish French and Russian, **how many students have taken a course in all 3 languages.**

Solution: Let S be the set of students who have taken a course in Spanish, F the set of students who have taken a course in French, and R the set of students who have taken a course in Russian. Then, we have

$|S| = 1232$, $|F| = 879$, $|R| = 114$, $|S \cap F| = 103$, $|S \cap R| = 23$, $|F \cap R| = 14$, and $|S \cup F \cup R| = 2092$.

Using the equation

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|,$$

we obtain $2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$.

Solving for $|S \cap F \cap R|$ yields 7.

The Principle of Inclusion-Exclusion

Theorem 1. The Principle of Inclusion-Exclusion:

Let $A_1, A_2, \dots, A_n$ be finite sets. Then:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \\ & \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \\ & \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

Applications of Inclusion-Exclusion

Section 8.6

The Number of Onto Functions

Example: How many **onto functions** are there from a set with 6 elements to a set with 3 elements?

Solution: Suppose that the elements in the codomain are b_1, b_2 , and b_3 . Let P_1 , P_2 , and P_3 be the properties that b_1 , b_2 , and b_3 are not in the range of the function, respectively. The function is onto if none of the properties P_1, P_2 , and P_3 hold.

By the **inclusion-exclusion principle** the number of onto functions from a set with 6 elements to a set with 3 elements is

$$N - [N(P_1) + N(P_2) + N(P_3)] + [N(P_1P_2) + N(P_1P_3) + N(P_2P_3)] - N(P_1P_2P_3)$$

- Here the total number of functions from a set with six elements to one with three elements is $N = 3^6$.
- The number of functions that do not have in the range is $N(P_1) = 2^6$. Similarly, $N(P_2) = N(P_3) = 2^6$.
- Note that $N(P_1P_2) = N(P_1P_3) = N(P_2P_3) = 1$ and $N(P_1P_2P_3) = 0$.

Hence, the number of onto functions from a set with six elements to a set with three elements is:

$$3^6 - 3 \cdot 2^6 + 3 = 729 - 192 + 3 = 540$$

The Number of Onto Functions (continued)

Theorem 1: Let m and n be positive integers with $m \geq n$.

Then there are

$$n^m - C(n, 1)(n - 1)^m + C(n, 2)(n - 2)^m - \dots + (-1)^{n-1}C(n, n - 1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Proof follows from the principle of inclusion-exclusion

Derangements (錯位)

Definition: A *derangement* is a permutation of objects that leaves no object in the original position.

Example: The permutation of 21453 is a derangement of 12345 because no number is left in its original position. But 21543 is not a derangement of 12345, because 4 is in its original position.

Derangements (continued)

Theorem 2: The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right].$$

Proof follows from the principle of inclusion-exclusion (*see text*).

Derangements (continued)

The Hatcheck Problem: A new employee checks the hats of n people at restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. **What is the probability that no one receives the correct hat.**

Solution: The answer is the number of ways the hats can be arranged so that there is no hat in its original position divided by $n!$, the number of permutations of n hats.

Remark: It can be shown that the probability of a derangement approaches $1/e$ as n grows without bound.

$$\frac{D_n}{n!} = \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right]$$

TABLE 1 The Probability of a Derangement.

n	2	3	4	5	6	7
$D_n/n!$	0.50000	0.33333	0.37500	0.36667	0.36806	0.36786